

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0412

Sturgis 100 kV
March 17, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0412. The Interconnection Customer proposes a 75.4 MW wind facility interconnecting at the Sturgis 100 kV substation in the Michigan area. The request was received on December 17, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 233 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,643,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
THREE RIVERS SUBSTATION - HYDRO PLANT 100.0 kV Line	PALISADES - OCEANSIDE 500.0 kV	111.4	151	7.4
RIDGELINE - HYDRO PLANT 100.0 kV	PALISADES - OCEANSIDE 500.0 kV	109.8	154	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
THREE RIVERS - HYDRO PLANT 100.0 kV	Rebuild line with higher-capacity conductor	\$4,887,000	100% (\$4,887,000)
RIDGELINE - HYDRO PLANT 100.0 kV	Reconductor line	\$3,756,000	100% (\$3,756,000)
	Total allocated to project		\$8,643,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.